

The test is *finite* if, for some  $n \in \mathbb{N}$ ,  $T$  depends only on the forecasts and outcomes of the first  $n$  periods.

**Definition 2.** Fix a prequential test  $T$  and let  $\mu$  be a theory.

- (1)  $\mu$  *passes the test over a realization*  $x = (s_0, s_1, \dots) \in S^{\mathbb{N}}$  if  $\mu(N(s_0, \dots, s_{n-1})) > 0$  for every  $n$  and  $T(p_0, s_0, p_1, s_1, \dots) = \text{PASS}$ , where  $p_k$  is the forecast of  $\mu$  over  $x$  at period  $k$ .
- (2) The *acceptance set of  $\mu$  under  $T$* , denoted  $A_\mu^{(T)}$  is the set of all realizations  $x \in S^{\mathbb{N}}$  such that  $\mu$  passes the test over  $x$ .

**Definition 3.** The prequential test  $T$  *accepts the data generating process in a paradigm  $\Gamma$  with probability  $1 - \epsilon$*  if  $\mu(A_\mu^{(T)}) \geq 1 - \epsilon$  for every  $\mu \in \Gamma$ .

If the test accepts the data generating process with high probability and Bob's theory  $\mu$  is the data generating process then Bob is assured that he will pass the test with high probability.

**Definition 4.** Let  $\Gamma$  be a paradigm. A test  $T$  is  $\epsilon$ -manipulable w.r.t  $\Gamma$  if there exists some distribution  $\zeta$  over  $\Gamma$  such that

$$\zeta \otimes \mu(\{(\nu, x) : x \in A_\nu^{(T)}\}) > 1 - \epsilon$$

for every theory  $\mu \in \Gamma$ .

If a test is  $\epsilon$ -manipulable then Bob can randomize a theory according to  $\zeta$  and be sure that, with high probability, he passes the test regardless of the data generating process  $\mu$ .

If a test is not manipulable then every strategy to fake theories fails on at least one data-generating process. Following Dekel and Feinberg (2006) and Olszewski and Sandroni (2009b), we use in this paper a stronger condition, which entails that every strategy employed by an ignorant expert will fail on a topologically large set of data generating processes. Recall that a set  $M$  of a separable metric space  $Z$  is co-meager if it contains a countable intersection